

Modelling of Large-Scale Circulation in Atmosphere and Ocean
(MO8004)

Kelvin Waves

Lab 2

May 19, 2026

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Questions

Theory

Coastal Kelvin Waves

1. Describe in your own words what coastal Kelvin waves are.
2. State the necessary assumptions to study coastal Kelvin waves.
3. Write down the solutions for the Shallow Water Equations for coastal Kelvin waves.
4. Describe the shape of the coastal Kelvin waves.

Equatorial Kelvin Waves

1. Describe the assumptions to study equatorial Kelvin waves.
2. Write down the solution for the Shallow Water Equations for equatorial Kelvin waves.
3. In which direction do the equatorial Kelvin waves travel?

Experiments

Set up the model as described in the laboratory description for Laboratory 2 and follow the instruction for the experiments.

Coastal Kelvin Waves

1. Compute the phase speed from the position of the peak of the wave along the x-axis at $y = 0$ and at different times.
2. Compare your results to the theory and discuss weaknesses of your results.
3. Provide plots showing the cross-section in the y-direction of your simulated coastal Kelvin waves. Ensure to include the analytical solution and the initial condition.
4. How is the result dependent on the resolution?
5. Show plots of the cross section of these waves in x-direction for your experiments at different time steps.
6. Discuss changes of the wave with time and with resolution.

Equatorial Kelvin Waves

1. Why do we want reduced gravity in this simulation?
2. Provide a plot of the horizontal cross-section at an appropriate time step and discuss the waves you can identify (include a cross-section through the equatorial Rossby waves in the same plot).
3. Provide a plot of the wave for multiple points in time (in x-direction, at $y = 0$).
4. From that, compute the phase speeds and compare to theory.
5. Provide a plot showing cross sections of the wave amplitudes in y-direction. Make sure to also include the initial condition and the analytical solution and compare your result to the theory.
6. Provide examples of where these waves are observed in the real ocean.

1 Theory

1.1 Coastal Kelvin Waves.

i) Kelvin waves are a geostrophic phenomenon where the presence of a coast forces an initial perturbation to propagate along that coast. The movement is anticlockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.

The reason for these waves propagating the way they do lies in the presence of the boundary (like a coast or the equator). In normal geostrophic balance a large-scale high or low (same as a 'bump' or 'valley') can exist due to geostrophic movement around the high/low, maintaining the shape due to the presence of the Coriolis force.

But if a boundary is present, no such flow around the high/low can exist. If one for example has a high, normal geostrophy tells us that we need anticyclonic motion around the high to keep the 'bump' in place. However, if we were to have a southern boundary (and we assume that we are in the Northern Hemisphere), then we would lack westerly geostrophic flow. Thus, due to the conservation of momentum, this high would have to propagate to the East. That is, in other words, a Kelvin wave.

ii) To study Kelvin waves one needs three critical assumptions; a set of geostrophic equations, some sort of solid boundary, and an initial perturbation. The set of equations can be any sort that includes time dependence and rotation. A common easy choice is to use the shallow water equations as

$$\begin{cases} \frac{\partial u}{\partial t} - fv = -g' \frac{\partial \eta}{\partial x} \\ \frac{\partial v}{\partial t} + fu = -g' \frac{\partial \eta}{\partial y} \\ \frac{\partial \eta}{\partial t} + H \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) = 0 \end{cases} \quad \begin{matrix} (1a) \\ (1b) \\ (1c) \end{matrix}$$

where u and v are the horizontal velocity components in the x and y directions. η is the free surface height, H is the mean fluid depth, f is the Coriolis parameter, and g' is the reduced gravity parameter defined as

$$g' \equiv \frac{\rho_2 - \rho_1}{\rho_1}.$$

The reduced gravity model assumes that we have a moving fluid layer of density ρ_1 on top of a non-moving layer of density ρ_2 . The lower layer is assumed to be a lot deeper.

The solid boundary can be something like a coast or the equator (where the Coriolis force is equal to zero). Thus the boundary is a solid boundary where the fluid cannot pass through. Finally, the initial perturbation has to be of an adequate size for the Coriolis force to be able to act upon it. A normal way of measuring this is stating that the perturbation has to be larger than the Rossby radius of deformation defined as

$$R \equiv \frac{\sqrt{g'H}}{f}$$

away from the equator, or near the equator as

$$R_{\text{equator}} \equiv \sqrt{\frac{\sqrt{g'H}}{\beta}} \text{ alternatively } R_{\text{equator}} \equiv \sqrt{\frac{\sqrt{g'H}}{2\beta}}.$$

Above, the non-equatorial Rossby radius uses the f-plane approximation where $f \approx f_0$, and the equatorial Rossby radius uses the beta-plane approximation as $f \approx f_0 + \beta y$, where $f_0 = 0$ at the equator.

iii) If one assumes that we have a southern boundary, like a coast and an initial perturbation larger than the Rossby radius of deformation, then we can derive an solution to the coastal Kelvin waves. The solution is

$$\begin{cases} \eta = \eta_0 (x - ct) e^{-\frac{y}{R}} & (2a) \\ u = \sqrt{\frac{g'}{H}} \eta_0 (x - ct) e^{-\frac{y}{R}} & (2b) \\ v = 0 & (2c) \end{cases}$$

where η_0 is the initial perturbation, and c is the non-dispersive wave phase speed equal to $\sqrt{g'H}$.

Furthermore, one must note that this solution is only for the Northern Hemisphere. If one would be located at the Southern Hemisphere, then the Rossby radius of deformation would essentially 'be negative', which would lead to exponential increase in amplitude away from the coast. To combat this, one can derive an identical solution with the only change being that there is no minus sign in front of the exponent, and that the propagation is in the other direction. I.e. $c \rightarrow -c$ and $\exp(-\frac{y}{R}) \rightarrow \exp(\frac{y}{R})$, where $R < 0$.

iv) Observing the solution one can observe that the solution is an exponential decay away from the coast. The decay is

$$\eta \propto e^{-y/R},$$

and the solution will thus keep its initial height along the coast. The zonal shape will stay as the initial shape as we keep the η_0 term. Thus, the solution will be a propagating wave as the initial condition in the zonal-direction, and decay exponentially away from the coast. On the N.H. the propagation will be with the coast to the right.

1.2 Equatorial Kelvin Waves

i) Essentially, equatorial Kelvin waves are just like coastal Kelvin waves, except that the boundary here is the equator. Since the Coriolis force is zero at the equator geostrophic flows cannot pass this line. Essentially this works the same way as described earlier, a large perturbation usually wants to stay in geostrophic balance, but since there cannot be any geostrophic flow along the equator, conservation of momentum will force the perturbation to move East.

The assumptions here are thus very similar to the coastal Kelvin waves, a set of geostrophic equations, an initial perturbation, and a close proximity to the equator. A suitable assumption to study this is to use the beta-plane approximation, which centred at the equator is

$$f \approx \beta y.$$

Above β is an always positive parameter, describing the linearized Coriolis term. The coordinate y is centered at the equator, meaning that a location on the Southern Hemisphere yields a negative y .

ii) If one assumes that we are close to the equator and that we have initial perturbation larger than the equatorial Rossby radius of deformation R_{eq} , then we can derive an solution to the equatorial coastal Kelvin waves. The solution is

$$\begin{cases} \eta = \eta_0 (x - ct) e^{-\frac{y^2}{2R_{eq}^2}} & (3a) \\ u = \sqrt{\frac{g'}{H}} \eta_0 (x - ct) e^{-\frac{y^2}{2R_{eq}^2}} & (3b) \\ v = 0 & (3c) \end{cases}$$

Note that there are two values of the equatorial Rossby radius of deformation, with a difference of a factor $\sqrt{\frac{1}{2}}$. In this report the definition without the $\sqrt{\frac{1}{2}}$ will be used.

iii) Since the exponential term in Equation 3 contains a y^2 , the exponent will always be negative. The solution is thus applicable to both the Southern and Northern Hemisphere. The solution will thus always propagate in the positive x -direction i.e. East. One could also explain this through the beta-plane

approximation, which shows that the β -term is always positive. One can visualize this as the two equatorial Kelvin waves 'leans' onto each other, and are each others coast. Thus the equatorial Kelvin wave will always propagate to the East.

2 Experiments

2.1 Coastal Kelvin Waves

The coastal Kelvin waves simulations used two different oceans as a baseline. The Atlantic ocean used a grid size of $1e7 \times 1e7$ m² and a mean depth of 1000 m. The Baltic sea used a grid size of $1e6 \times 1e6$ m² and a mean depth of 30 m. The initial perturbation was in both cases a gaussian perturbation with a top height of 5 m and a extent of 10% of the domain. The perturbation was placed in the middle of the southern border, and can be seen in Figure 1. Both oceans were also tested on two different grids, one with a grid size of 200×200 and a finer one with 600×600 .

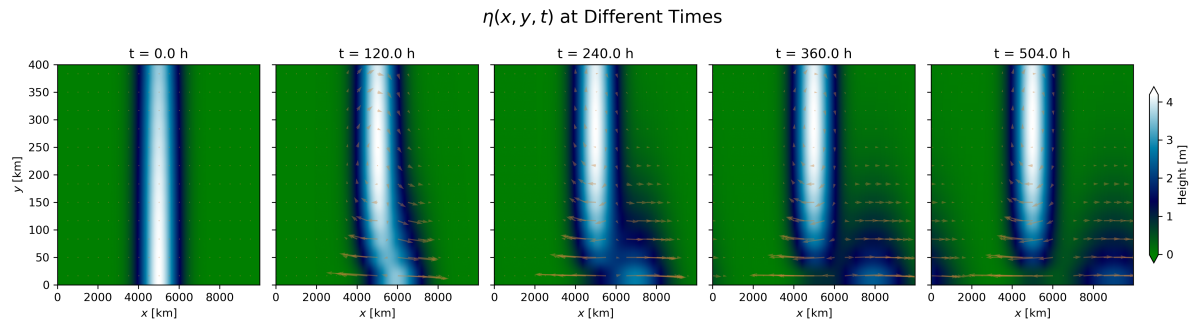


Figure 1: This figure shows the time evolution of the initial perturbation along the southern border.

i,ii) Calculating the phase speed of the different schemes, between different times, can be seen in Table 1. There one can note a small, but present, difference between the numerical and theoretical phase speeds, where the numerical is always somewhat slower. The reason for this is due to the difference between a discretized computational phase speed and a theoretical one. Since the model uses a discretized staggered leapfrog scheme, the model is not identical to the theoretical equations and the numerical phase speed and the theoretical one are not the same.

Case	Time (h)	c_{num}	c_{theo}	Rel. error (%)
Atlantic	0 → 10	10.7639	9.9045	8.68
	0 → 24	8.5359	9.9045	13.82
	0 → 168	9.7760	9.9045	1.30
	0 → 336	9.8070	9.9045	0.99
	0 → 504	9.8035	9.9045	1.02
Atlantic fine grid	0 → 10	11.4583	9.9045	15.69
	0 → 24	9.5968	9.9045	3.11
	0 → 168	9.8173	9.9045	0.88
	0 → 336	9.8139	9.9045	0.92
	0 → 504	9.8173	9.9045	0.88
Baltic	0 → 10	1.9097	1.7155	11.32
	0 → 24	1.7216	1.7155	0.36
	0 → 168	1.7051	1.7155	0.61
	0 → 336	1.7061	1.7155	0.55
	0 → 504	1.7092	1.7155	0.37
Baltic fine grid 0 → 10	2.0255	1.7155	18.07	
	0 → 24	1.7120	1.7155	0.21
	0 → 168	1.7120	1.7155	0.21
	0 → 336	1.7123	1.7155	0.19
	0 → 504	1.7134	1.7155	0.12

Table 1: Coastal Kelvin wave phase speeds.

According to Döös et al. [2022] the theoretical and numerical phase speed for the shallow water equations discretized on an Arakawa C-grid are related as

$$\frac{c_{num}}{c_{theo}} = \pm \frac{1}{k\mu\Delta x} \arcsin \left[2\mu \sin \left(\frac{k\Delta x}{2} \right) \right]. \quad (4)$$

Above μ is the Courant number, k is the wave number and $\Delta x = \Delta y$ is the size of each grid box. Thus, by increasing the grid size, the ratio of $\frac{c_{num}}{c_{theo}} \rightarrow 1$. This can be seen for the two different oceans in Figure 2. One can observe that the Baltic sea converges faster than the Atlantic. This can be explained by Equation 4 and the Courant number defined as

$$\mu \equiv c \frac{\Delta x}{\Delta t}.$$

Since the Baltic sea is a smaller sea, the same grid size corresponds to smaller grid boxes, i.e. a smaller Courant number, and thus a better computational phase speed. This also explains why increasing the grid size for the Baltic sea in Table 1 led to a more accurate result.

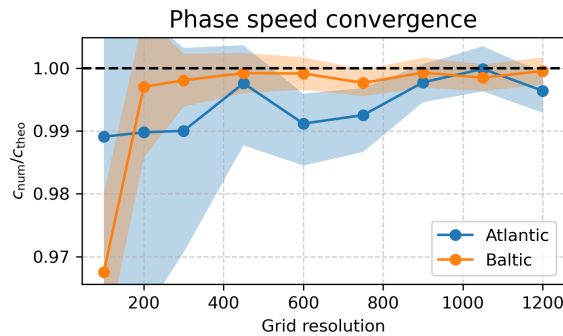


Figure 2: This figure shows the ratio between the numerical and theoretical phase speed for the two oceans for different grid sizes. The shaded regions show the spread of the phase speeds by one standard deviation. The actual points are the phase speed calculated between the hour 0 and 504.

iii) The meridional cross-section of the solution with the initial condition and theoretical solution can be seen in Figure 3. The theoretical solution is labelled $\eta_{theo, modified}$ since the computational phase speed

difference had been accounted for. Furthermore, since the solution also dropped in initial height (as can be seen in both Figure 1 and Figure 5), the relative height decrease from the maximum h -value is shown.

One interesting observation that can be seen here in Figure 3 but also in Figure 1, is that the initial perturbation is split up into an geostrophic-high part and a coastal Kelvin wave part. Thus, as only the coastal Kelvin wave part propagates, it needs to propagate away from the geostrophic-high for us to be able to analyze the exponential decay of the Kelvin wave. In Figure 3, the left most column shows the initial stage, where both parts are present. The middle column shows a stage where parts of the Kelvin wave are past the geostrophic-high.

However, in the rightmost column one can observe when the Kelvin waves has propagated away from the geostrophic high. Comparing this with the expected theoretical decay of

$$\eta \propto e^{-\frac{y}{R}}$$

shows a very good agreement with the expected output. Thus the numerical models behave as expected in their decay from the coast.

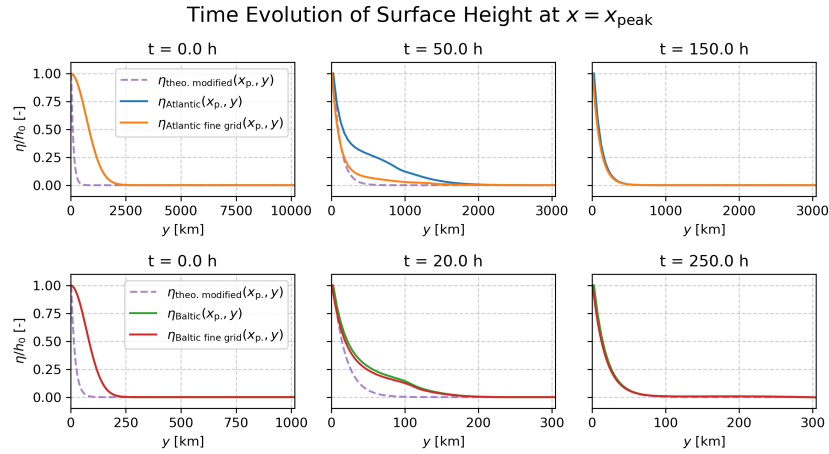


Figure 3: This figure shows the decay of the numerical solutions from the coast together with the initial conditions (left column) and the modified theoretical solution. The middle column show the decay when the Kelvin waves has not yet fully propagated away from the geostrophic high, and the rightmost column show when the Kelvin waves have propagated away from the geostrophic high.

iv) The dependence on the resolution in Figure 3 is very hard to see since the change is very close to the wave compared to the entire domain. One can thus look very close to the wave, and also account for the smallest length scale of the domain. Doing this for many different grid sizes gives Figure 4, where one can see that the decay is actually faster than the theoretical one. This can be explained by the presence of small scale dissipation in the model, that increases the speed of the decay. Thus higher resolution gives more accurate decay.

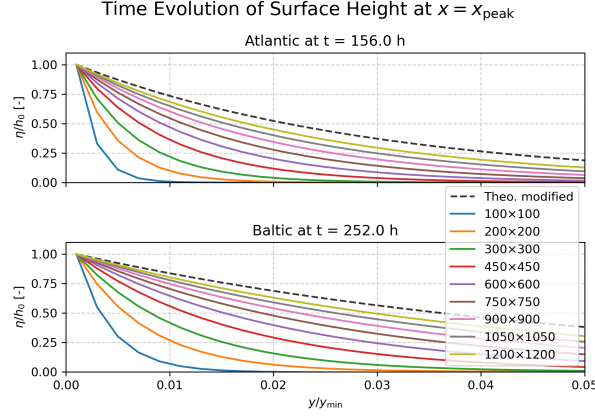


Figure 4: This figure shows the decay of the numerical solutions from the coast for a both of different grid sizes. Note that the x -axis is a normalized quantity over the min(L_x) to highlight the decay differences.

v) The zonal cross-section at the coast can be seen in Figure 5 together with the theoretical solution. This shows a gaussian shape initial perturbation that moves to the East with time. Furthermore, the shape seems to loose some height as time passes on.

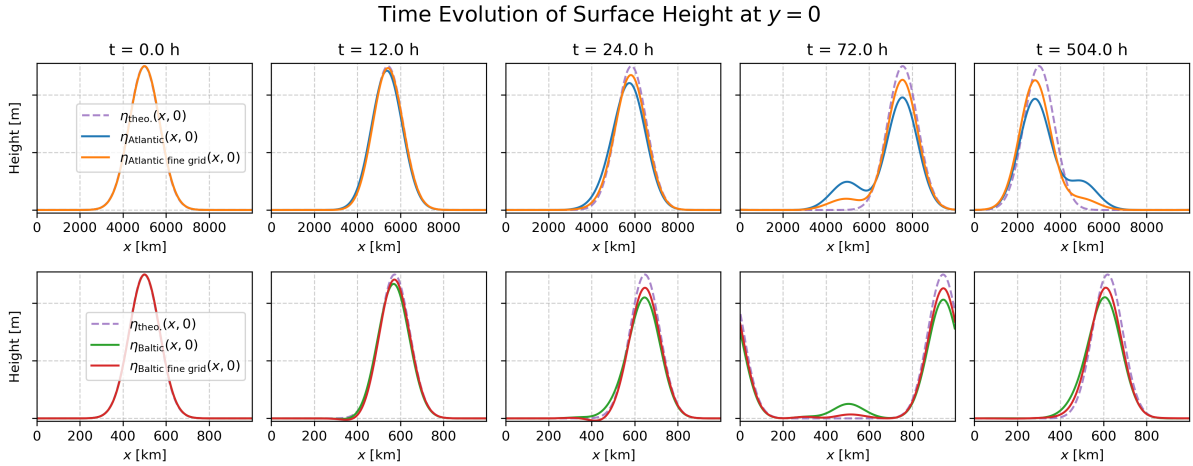


Figure 5: This figure shows the zonal propagation of the Kelvin waves in the two oceans. The theoretical solution is showed as a dotted line, and the initial conditions can also be seen.

vi) The change of the numerical Kelvin wave for different resolutions can be seen in Figure 6. There one can see that even the courser grids do a pretty good job of mirroring the theoretical solution. However there are some minor differences. As shown earlier in Table 1 one could see that the phase speed of the wave was slightly worsened for the courser grids, which can also be seen in the figure below. However, maybe more interestingly is that the courser grids lack more in height then the finer grids. Also, for the courser grid in the Atlantic some weird behaviour can be seen East of the wave. However these are not shown for the finer grids.

Interestingly, even the finer grids do not reach the same height as the theoretical Kelvin wave. This may be for two reasons. Firstly, the model shows numerical dissipation that slowly dampens the solution with time. Another reason may be dampening of factors such as the Robert-Asselin filter, which is known to slowly dampen the solution [Döös et al., 2022]. This is connected to this three level scheme having a computational mode, which affects the solution.

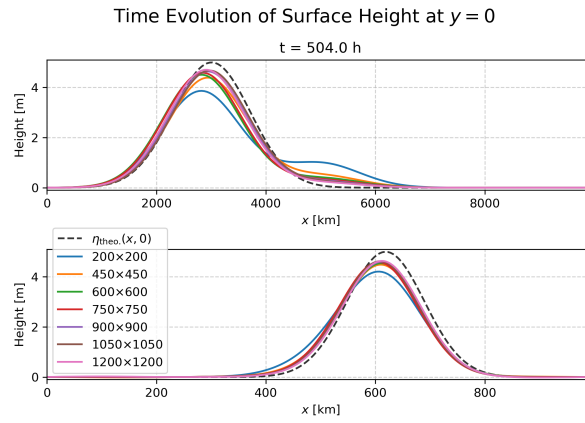


Figure 6: This figure shows the zonal propagation of the Kelvin waves in the two oceans for a bunch of different grid sizes.

2.2 Equatorial Kelvin Waves

(i) Reduced gravity is an essential tool to get more physical like solutions. Previously, the Rossby radius of deformation was here defined as

$$R_{\text{equator}} \equiv \sqrt{\frac{\sqrt{g'H}}{\beta}}.$$

If one was to compute the Rossby radius of deformation with some common parameters such as $H = 1000 \text{ m}$, $\beta = 2 \times 10^{-11} \text{ m}^{-1} \text{ s}$ and a normal gravity of $g = 9.82 \text{ m s}^{-2}$, one would obtain a Rossby radius of more then 2000 km, which is more then the distance from Stockholm to Istanbul! Thus this is a highly unrealistic value. If one instead uses a reduced gravity of $g' = 0.0981 \text{ m s}^{-2}$ one gets a large radius, but far more realistic.

ii) A horizontal cross-section plot of the equatorial Kelvin wave can be seen in Figure 7. There one can see two things happening. Something propagating as a Gaussian shape to the East, which has to be the equatorial Kelvin wave since this aligns with the theory. However, two other objects are shown, to propagate more slowly to the west. This has to be the equatorial Rossby wave, since these are known to propagate in the Western direction. Furthermore, it make sense that we have a westerly propagating equatorial Rossby wave since this is the only way to preserve momentum.

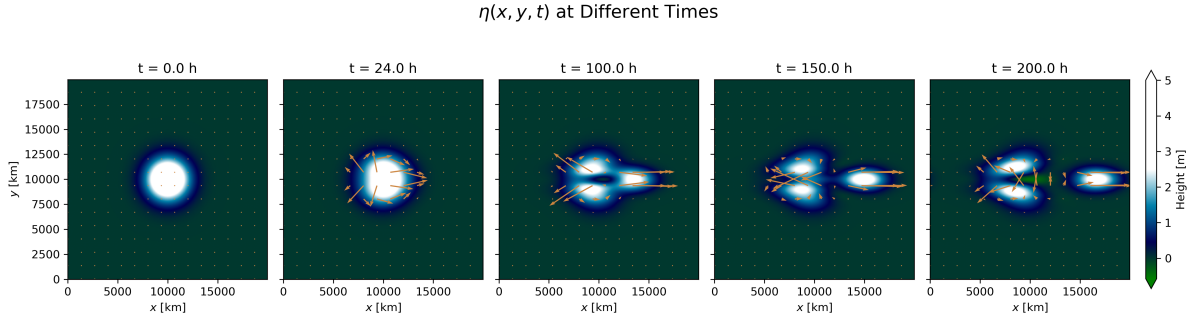


Figure 7: This figure shows the horizontal cross-section of the equatorial Kelvin wave and equatorial Rossby waves. The orange arrows indicate the velocity. This model was created by a grid size of $2e7 \times 2e7 \text{ m}^2$ with a resolution of 300×300 . The mean depth is 1000 m, and the initial perturbation is as before.

iii) A plot of the wave at along the equator ca be seen in Figure 8. There can one also see the initial conditions and the theoretical solution. One can also observe that the Kelvin wave has a smaller amplitude then expected. This may be due to the fact that some of the volume is carried away by the Rossby waves, which results in a smaller Kelvin wave.

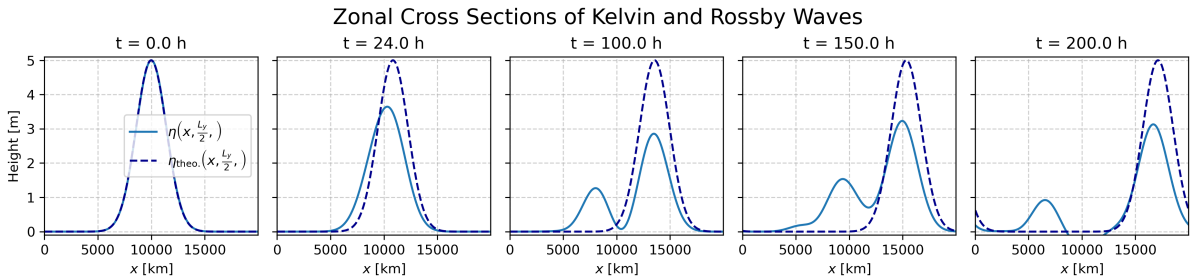


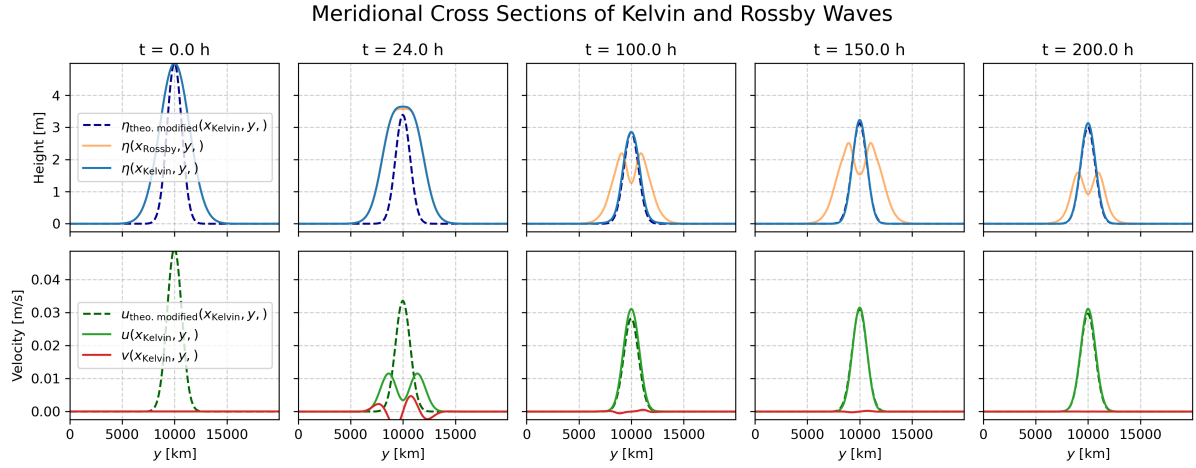
Figure 8: This figure shows the equatorial Kelvin wave for different times along the equator.

iv) From Figure 8 the phase speed can be computed, which can be seen in Table 2. There one can see similar phase speed accuracy to the coastal Kelvin wave, with a accuracy of 6% after 200 h. The reason for the slower moving computational model is the same reason as before (see Equation 4). The discretized version is never the same as the theoretical version, which gives an error in the phase speed and a slower phase speed.

Time [h]	$x_{\max}$ [km]	c_{num} [m/s]	c_{theo} [m/s]	Error [%]
0 $\rightarrow$ 24	10300.0	3.86	9.90	61.05
24 $\rightarrow$ 100	13500.0	11.70	9.90	18.09
100 $\rightarrow$ 150	14966.7	8.15	9.90	17.73
150 $\rightarrow$ 200	16700.0	9.63	9.90	2.78
0 $\rightarrow$ 200	16700.0	9.33	9.90	5.81

Table 2: Equatorial Kelvin-wave phase speed diagnostics taken from Figure 8.

v) The meridional cross-sectional amplitude decay of the numerical solution can be seen in Figure 9. This is shown together with the initial condition, the theoretical solution and the equatorial Rossby wave. There one can observe that the initial state is split up between the equatorial Kelvin wave and the equatorial Rossby wave. Since the initial perturbation is split up, the theoretical solution has been modified to use the same amplitude as the equatorial Kelvin wave to provide insight into the decay of the solution. There one can see that after around 100 hours, the theoretical decay solutions agrees very well with the numerical decay. The same can be said for the velocity, which also agrees well with the theoretical solution. This theoretical solution used the Rossby radius of deformation defined without the $\sqrt{\frac{1}{2}}$ -factor. Thus one can from this deduce that the Rossby radius without the $\sqrt{\frac{1}{2}}$ -factor works well.

**Figure 9:** This figure shows the cross-sectional solution of the equatorial Kelvin wave and the equatorial Rossby wave, at the location of the peak of the wave.

vi) Equatorial Kelvin waves are primarily associated with the Pacific Ocean, and especially the El Niño Southern Oscillation event. During an El Niño, the Walker circulation feeds on itself, creating a large wind stress over the Pacific. This creates higher water height near the western boundary, compared to the eastern boundary. Thus, the equatorial Kelvin wave is an important way of relaxation of El Niño, as it transports energy eastward!

These waves can often be measured and found by observing the wind stress over the Pacific Ocean. More interestingly, these equatorial Kelvin waves are known to create coastal Kelvin waves when they hit the South American coast, which connects the two types of Kelvin waves studied here. The same type of El Niño-like events does not occur in the Atlantic due to that ocean being much smaller than the Pacific.

A The Source Code

The source code for this lab can be found on GitHub:

`https://github.com/FredrikBergelv/
Modelling-of-Large-Scale-Circulation-in-Atmosphere-and-Ocean`

References

Kristofer Döös, Peter Lundberg, and Aitor Aldama Campino. *Basic Numerical Methods in Meteorology and Oceanography*. Stockholm University Press, Stockholm, Sweden, 2022. ISBN 978-91-7635-172-7. doi: 10.16993/bbs.